

A SIGNED-DIRECTION INVARIANT FOR TRIANGLE TILINGS, AND THE EXCLUSION OF PRIMES CONGRUENT TO 3 MODULO 4

VICO BONFIOLI

ABSTRACT. For a triangle T and a triangle ABC , an N -tiling is a dissection of ABC into N triangles each congruent to T . Erdős asked which N occur; a folklore conjecture is that no prime $N \equiv 3 \pmod{4}$ with $N > 3$ does (the value $N = 3$ occurs). By the classification of Laczkovich and the branch theorems of Beeson, the conjecture reduces to a tile with a $2\pi/3$ angle on a non-equilateral triangle, where a prime count forces ABC isosceles. For that isosceles case Beeson ruled out $N < 36$ and explicitly left open whether N can be prime (the smallest *known* tiling has 2673 tiles); we settle it. We introduce a translation-invariant, signed-direction tiling functional, linear in edge length and hence insensitive to non-edge-to-edge incidences, whose integrality forces, for an isosceles target, that $(c - a - b)/\sqrt{b}$ be an integer for a primitive triple with $c^2 = a^2 + ab + b^2$; an elementary lemma shows this never occurs. With a self-contained reduction of the scalene shapes, this completes the exclusion of primes $\equiv 3 \pmod{4}$ exceeding 3, conditional on the cited classification of the remaining branches. Consequently the prime case of the full problem is complete: a prime p occurs if and only if $p = 2$, $p = 3$, or $p \equiv 1 \pmod{4}$. Beyond primes, we determine the admissible spectrum of each sporadic $2\pi/3$ branch — for the isosceles target, writing $b = de^2$ with d squarefree, every count has the form $N = dw^2(a + 2b)$ with $e \mid w(c - a - b)$ — bounding the realizable set from outside; and a spectrum theorem shows that the counts passing all invariant conditions on the isosceles and F_1 targets are *exactly* Zhang’s constructed families, closing the necessary side of those branches; the equilateral criteria likewise reduce to elementary divisor conditions. Combining the spectra with the published branch theorems and an exhaustive exact search of the surviving finite instances, we settle every previously undetermined value up to 80 with one pending exception: no triangle can be cut into 14, 15, 21, 22, 30, 33, 35, 38, 39, 42, 51, 55, 56, 57, 60, 63, 69, or 76 congruent triangles (46, 62, 78 likewise, and 79 — a prime — by the main theorem); triangles *can* be cut into 44, 77 (by Beeson’s four-component construction), and 80 (a sum of two squares) congruent triangles; the values 66 and 70, previously excluded via theorems of [2] whose proofs we show to be unsound, are undetermined pending direct exhaustive searches of their single surviving instances (in progress); and *a triangle can be cut into 44 congruent triangles* — the isosceles (16, 16, 22) into 44 copies of the (2, 3, 4) tile, found by the search and verified exactly, to our knowledge the first tiling in the isosceles $3\alpha + 2\beta = \pi$ cases and the smallest known tiling in any incommensurable branch. Membership in the tile-count set is shown to be decidable. The arithmetic and combinatorial layer is machine-checked in Lean 4 with Mathlib.

1. INTRODUCTION

Let T be a triangle (the *tile*). An N -tiling of a triangle ABC is a way of writing ABC as a union of N triangles, each congruent to T , with pairwise disjoint interiors. We do *not* assume the tiling is edge-to-edge: a vertex of one tile may lie in the relative interior of an edge of another. Erdős asked (reported by Soifer [13]) to determine the set of N for which some triangle admits an N -tiling by some tile. Every perfect square occurs (any triangle, by the n^2 subdivision), and Soifer observed that $n^2 + m^2$, $2n^2$, $3n^2$ and $6n^2$ occur as well; in particular $3 = 3 \cdot 1^2$ occurs. The outstanding question is whether the primes $\equiv 3 \pmod{4}$ that exceed 3, which are exactly the odd primes that are neither sums of two squares nor of the form $3n^2$, are excluded.

Theorem 1. *No prime $N \equiv 3 \pmod{4}$ with $N > 3$ is a number of congruent triangles into which some triangle can be cut. In particular, no triangle can be cut into 19 congruent triangles.*

The restriction $N > 3$ is necessary: an equilateral triangle joined to its centre splits into 3 congruent triangles, so $N = 3$ occurs.

The proof rests on Laczkovich’s classification of triangle tilings [9, 10], organized by Beeson across a series of papers [1, 2, 3, 4, 5]. For a prime number of tiles every branch of the classification is excluded by an existing theorem, with one exception: a tile having a $2\pi/3$ angle, tiling a non-equilateral triangle. We treat that branch.

This branch is the active frontier. Zhang [7] classifies the non-equilateral $2\pi/3$ shapes into six sporadic similarity types (Section 2 below), constructs families of tilings for each, and conjectures that the constructed N , all composite, are the only ones; the present results prove the no-prime half of that conjecture for the isosceles type, and reduce the scalene types directly. For the isosceles type Beeson [3] proved that no tiling exists with $N < 36$ and *explicitly left open whether N can be prime*; the smallest tiling presently known has 2673 tiles (Herdt, reported in [3]). The invariant below settles the question, excluding *every* prime. The adjacent Erdős Problem 633, on tilings into a square number of triangles, was recently settled by Beeson, Laczkovich and Zhang [8].

Two reductions prepare it. First, such a tile has commensurable sides, so after scaling we may take it integer-sided with $c^2 = a^2 + ab + b^2$; for the *isosceles* targets to which the theorem reduces, this is Beeson’s *accepted* isosceles paper ([3, Thm. 12.5]: any isosceles triangle tiled by a $2\pi/3$ -tile with α/π irrational forces the tile rational), so the prime theorem’s rationality input is refereed; the general-tile rationality of Beeson–Zhang [6] gives the same uniformly. Second, an elementary area-and-incidence argument shows that a *prime* number of such tiles forces ABC to be *isosceles* (Section 3).

The heart of the paper is Section 4. The natural parity argument for a $2\pi/3$ tile (a black/white colouring of the tiles) fails, because a vertex of the tiling may be surrounded by an even number of tiles, so no consistent two-colouring exists. We replace it by a functional Φ that assigns to a directed edge of direction $\theta = j\frac{\pi}{3} + k\alpha$ the weight (length) $\cdot (-1)^j$. Two features make it work: $\Phi(\theta + \pi) = -\Phi(\theta)$, so interior edges cancel; and the weight is *linear in edge length*, so the cancellation is unaffected by non-edge-to-edge incidences, where one long edge meets several shorter ones summing to it. Summing over the tiles, Φ of the boundary equals an integer multiple of a fixed tile-value $c + a - b$. For an isosceles target this forces $(c - a - b)/\sqrt{b} \in \mathbb{Z}$, which never holds (Section 5). Assembling the branches proves Theorem 1 (Section 6).

The combinatorial claims are verified by the scripts in the accompanying repository, and the entire arithmetic layer, including Lemma 15 and the reduction of Section 3, is formalized and machine-checked in Lean 4 with Mathlib; see Section 8.

2. PRELIMINARIES

The classification. Fix a prime $N = p > 3$ with $p \equiv 3 \pmod{4}$. By Laczkovich [9, 10], every N -tiling of a triangle falls into one of finitely many types of pairs (ABC, T) . The relevant dichotomy, in Beeson’s organization [4], is: either the angles of T are pairwise commensurable with π , or they are not. If they are commensurable, then N is a square, a sum of two squares, or 2, 3 or 6 times a square [4, Thm. 3 and Table 3]; but $p > 3$ with $p \equiv 3 \pmod{4}$ is a square only if p is one (it is not), a sum of two squares only if $p \not\equiv 3 \pmod{4}$, and $2n^2, 3n^2, 6n^2$ only if p is even or $3 \mid p$ (here p is odd and $p > 3$), so p is none of these (it is precisely for $p = 3 = 3 \cdot 1^2$ that this step would fail). If the angles are not commensurable, then, after the right-angle and similar-tile cases, which give N a sum of two squares or $N = n^2$ [1, 12], the tile has a special angle and falls into one of: $3\alpha + 2\beta = \pi$, for which N is not prime (originally [2]; see Remark 2

and Propositions 10 and 3 — the arithmetic core is machine-checked in Lean for *four* of the five target shapes, with our own proofs replacing Beeson’s Theorems 18 and 20, whose printed arguments are unsound; the fifth, the base- β isosceles target, is the exception and is discussed honestly in Remark 4; an isosceles tile with $\gamma = 2\alpha$, for which N is not squarefree [3, Thm. 11.7]; or the tile has an angle equal to $\pi/3$ or $2\pi/3$. The case of a $\pi/3$ or $2\pi/3$ tile on an *equilateral* triangle gives N not prime [5]. What remains for a prime N is a tile with a $\pi/3$ or $2\pi/3$ angle on a *non-equilateral* triangle. The vertex enumeration of Section 2 shows the $\pi/3$ case admits only equilateral or tile-similar ABC , already covered; so the sole remaining branch is a $2\pi/3$ tile, which we now fix.

Remark 2 (The $3\alpha + 2\beta = \pi$ prime exclusion). The step above invokes [2] for the claim that a $3\alpha + 2\beta = \pi$ tiling has N composite. For the two scalene targets $(2\alpha, \beta, \alpha + \beta)$ and $(2\alpha, \alpha, 2\beta)$ this is Beeson’s Theorems 8 and 12, whose arithmetic cores we machine-check in Lean (`lean/Beeson3NotPrime.lean`): the first tiling equation $N + M^2 = 2K^2$ with $K \mid N$ admits no odd prime, and the second forces $N = (2f^2 - e^2)(3f^2 - e^2)k^2$ — a product of two factors exceeding 1, hence composite. The isosceles target with base angles $\alpha + \beta$ is Beeson’s Theorem 18, but its published proof is *unsound*: under Beeson’s scaling $a = N - M^2$, $c = N + M^2$, $b = c - a^2/c$ one has $bc = c^2 - a^2 = 4NM^2$, so $c^2b = 4NM^2(N + M^2) \equiv 0 \pmod{N}$ *identically* (no primality used), not the $M^4(M^2 + 1)$ his argument needs — a dropped factor of a ; the residue is already false on genuine composite tilings ($N = 48$, $M = 4$ gives $c^2b \equiv 0$ while $M^4(M^2 + 1) \equiv 32$), so the step forcing $N = M^2 + 1$ is vacuous. The conclusion is nonetheless correct, by the base-length obstruction of Proposition 10.

Two further defects of [2] affect the base- α target. First, the squarefree assertion of its Lemma 8 (“ $g = \gcd(a, c)$ is squarefree”) is *false*: the tile $(4, 15, 16)$ — a genuine $3\alpha + 2\beta$ tile, $e = 1$, $f = 4$ — has $g = 4$ (its proof breaks at the inference $p^2 \mid a \Rightarrow p^2 \mid \hat{a}$); the parametrized family $(ef, f^2 - e^2, f^2)$ realizes *every* f coprime to e , squarefree or not. Second, Theorem 19’s claim $g \mid M$ rests on that lemma *and* on the assertion that $bc^3(a + c)$ is not divisible by g^5 — but with $c = g^2$ it carries g^7 identically, so the argument collapses, and with it the proof of the base- α prime exclusion (Theorem 20). Proposition 3 repairs the latter without any appeal to $g \mid M$.

Proposition 3 (Base- α prime exclusion, replacing [2, Thm. 20]). *No isosceles triangle with base angles α , where $3\alpha + 2\beta = \pi$ and $\alpha \notin \mathbb{Q}\pi$, is N -tiled for N prime.*

Proof. Write $s = e/f$ in lowest terms, so the tile is $(ef, f^2 - e^2, f^2)$. Two necessary conditions suffice: the tiling equation $N(f - e)(2f + e)^2 = M^2(f + e)(2f^2 - e^2)$ (the sound half of [2, Thm. 19]) and the integrality of the equal side X , where $X^2 = Nbc/(2 - s^2)$. Substituting the tiling equation into X^2 gives the exact value $X = M(f + e)f^2/(2f + e)$; as $\gcd(2f + e, f) = \gcd(2f + e, f + e) = 1$, integrality forces $(2f + e) \mid M$. Writing $M = (2f + e)m$ reduces the tiling equation to

$$N(f - e) = m^2(f + e)(2f^2 - e^2).$$

Put $d = f - e$ and $Q = 2f^2 - e^2 = e^2 + 4ed + 2d^2$. Then $\gcd(d, Q) = 1$ (a common prime would divide e^2 , hence e , hence $\gcd(e, f)$), so the minimal prime factor of $Q \geq 7$ divides N and equals N ; writing $Q = NQ_1$ and cancelling N leaves $d = m^2(f + e)Q_1 \geq f + e > f - e = d$, a contradiction. Both arithmetic pillars are machine-checked in Lean (`lean/IsoAlphaPrime.lean`, `axiom-clean: isoalpha.X_forces, isoalpha_not_prime`). $\square$

Remark 4 (The base- β target: an honest gap). The fifth target, isosceles with base angles β , is the one shape where we cannot replace Beeson’s argument with a sound one. Beeson’s Theorem 14 also asserts $g \mid M$, by a proof of the same broken shape as Theorem 19’s (it claims abc^5M^2 is not divisible by g^7 , but with $c = g^2$ it carries g^{11}). And here the defect *bites*: the

sound necessary conditions do not exclude prime N . With $s = e/f$ in lowest terms the tile is $(ef, f^2 - e^2, f^2)$, the equal side is $X = f^3 M / (f + e)$ and the base is $Y = eM(3f^2 - e^2) / (f + e)$; integrality of X forces $(f + e) \mid M$, and writing $M = (f + e)m$ the Theorem-14 equation $N(e + f)^2 = M^2(3f^2 - e^2)$ collapses to

$$N = m^2(3f^2 - e^2),$$

which is prime for $m = 1$ and $3f^2 - e^2$ prime (e.g. $N = 11, 23, 47, 59, 71, \dots$, always $\equiv 3 \pmod{4}$). Each such N passes every sound necessary condition ($X, Y \in \mathbb{Z}$, parity, range) and is killed only by the unsound $g \mid M$. Unlike the base- α case (Proposition 3), no arithmetic contradiction follows; *a general proof that these candidates do not tile is open*, and to our knowledge is not supplied by a correct argument in the literature. For the finitely many instances that bear on the values settled in this paper we resort to the direct corner-anchored search. An exhaustive audit of $N \leq 80$ — computing the equal side X and base Y exactly and retaining every candidate that passes all *sound* necessary conditions ($X, Y \in \mathbb{Z}$, parity, range) — shows the affected set is precisely

primes $\{11, 23, 47, 59, 71\}$ (base- β) and composites $\{21, 70\}$ (base- α), $\{39, 66\}$ (base- β).

The engine settles them one at a time; all searches are exact and exhaustive:

N	branch	tile	target	nodes
11	base- β	(2, 3, 4)	(8, 8, 11)	135
21	base- α	(2, 3, 4)	(12, 12, 21)	13 659
23	base- β	(6, 5, 9)	(27, 27, 46)	19 677
47	base- β	(4, 15, 16)	(64, 64, 47)	12 440
39	base- β	(12, 7, 16)	(64, 64, 117)	825 724
71	base- β	(10, 21, 25)	(125, 125, 142)	553 417

each returning EXHAUSTED, NO TILING. The remaining three — $N = 59$ (base- β , (20, 9, 25) on (125, 125, 236)), $N = 66$ (base- β , (15, 16, 25) on (125, 125, 198)) and $N = 70$ (base- α , Remark 33) — are still searching, and their values are treated as pending. We flag this plainly: for the base- β target our $3\alpha + 2\beta$ prime argument rests on either [2, Thm. 14] (whose printed proof is unsound) or a per-value search, *not* on a machine-checked general theorem — the one genuine gap remaining in the prime dichotomy — though Theorem 5 below now removes an infinite sub-family of it, and Remark 7 explains why the method [2] used cannot repair the rest.

Theorem 5 (The base- β family at $e = 1$). *Let $f \geq 2$ and let the tile be $(a, b, c) = (f, f^2 - 1, f^2)$ (the primitive $3\alpha + 2\beta = \pi$ tile at $e = 1$). Then the isosceles triangle with base angles β and sides $(X, X, Y) = (f^3, f^3, 3f^2 - 1)$ admits no tiling by $N = 3f^2 - 1$ copies of the tile, in each of the following cases:*

- (i) $f = 2$ (i.e. $N = 11$) — unconditionally;
- (ii) $f \geq 3$ — conditionally on [2, Lem. 6] (“every side of ABC carries at least two c -edges”), which we have not verified and which we flag as a real risk in Remark 6.

In case (ii) the family is $N = 3f^2 - 1 = 26, 47, 74, 107, 146, 191, 242, \dots$, and for each such N the triple $(e, f, m) = (1, f, 1)$ is the only solution of $N = m^2(3f^2 - e^2)$ with $1 \leq e < f$, $\gcd(e, f) = 1$, so the whole base- β sub-branch closes for those N — conditionally. Independently, $N = 11, 26, 47$ are confirmed by exhaustive search.

Proof sketch. Write $\omega = \alpha/2$. The law of cosines gives $\cos \alpha = (2f^2 - e^2)/(2f^2)$, hence $\sin \omega = e/(2f)$; as $1 \leq e < f$ places this strictly in $(0, \frac{1}{2})$, Niven’s theorem forces α/π irrational — for every (e, f) , with no appeal to [9] (tile_alpha_irrational). Irrationality pins the vertex figures: a vertex $x\alpha + y\beta + z\gamma = k\pi$ forces $y + z = 2k$ and $2x + z = 3y$, so a π -vertex is $\{3\alpha, 2\beta\}$ or $\{\alpha, \beta, \gamma\}$, each base corner carries *exactly one* tile (its β -vertex) and the apex

exactly three (`vertex_pi`, `vertex_beta_corner`, `vertex_apex`). Each side of ABC is partitioned into whole tile edges (a supporting line meets a convex tile in a face). At $e = 1$, $m = 1$ the base $Y = 3f^2 - 1$ then admits, among coverings with at least two c -edges, the unique multiset $\{b, c, c\}$ (`base_composition_e1`); since the corner tile's β -vertex is flanked by a and c and $a \notin \{b, c\}$, the base reads c, b, c and each corner tile puts its a -edge on the equal side. Let P_1, P_3 be the interior base division points and T_2 the tile on the middle b -edge; T_2 carries α at exactly one of P_1, P_3 and γ at the other. But the corner tile's own b -edge has both endpoints on ∂ABC and is transversal there, so by the same edge arithmetic mod f it is matched by a *single* tile T^* whose b -edge is exactly that segment; the type at P_1 cannot be $\{\alpha, \beta, \gamma\}$ (none of the three tiles meeting there can carry β), hence is $\{3\alpha, 2\beta\}$, which contains no γ — forcing T_2 to carry α at P_1 . The mirror argument at P_3 forces α there too, a contradiction.

The four arithmetic/analytic pillars (irrationality via Niven, the three vertex enumerations, and the base composition) are machine-checked in Lean (`lean/BaseBetaE1.lean`, axiom-clean). The argument is sharp: it does *not* touch the genuine 44-tiling, which is this same family at $m = 2$ — there the base walk is a, a, a, a, c, c, c, a and carries *no* b -edge, so the driver of the contradiction is absent. $\square$

Remark 6 (Exactly where the $e = 1$ argument is conditional). The step “the base is $\{b, c, c\}$ ” is where [2, Lem. 6] enters, and it is load-bearing. At $e = 1$, $m = 1$ the congruence $Q \equiv 1 \pmod{f}$ and the length $Y = 3f^2 - 1$ leave *three* compositions, not one:

$$\{b, c, c\}, \quad \{a^f, b, c\}, \quad \{a^{2f}, b\}.$$

A γ -trap argument (no boundary junction carries γ on both sides, so only a c -edge can leave the γ -state) shows citation-free that every side carries *at least one* c -edge, killing $\{a^{2f}, b\}$; and for $f = 2$ the corner-chord forcing kills the rest, giving case (i) unconditionally. But $\{a^f, b, c\}$ — e.g. the word $acbaa$ at $f = 3$, of length $3 + 9 + 8 + 3 + 3 = 26 = Y$ — survives every citation-free constraint we have; only Lemma 6's “at least *two* c -edges” removes it. We flag this prominently: [2] is the paper whose Lemma 8 we show to be *false* (Remark 2) and whose Theorems 14, 18, 19, 20 we show to be unsound. We therefore do not regard case (ii) as settled, and record $N = 107, 191, \dots$ (the $f \geq 5$ primes of this family) as resting on an unverified citation rather than on a proof of ours.

Remark 7 (Why the colouring method cannot close the rest). The residual gap ($e \geq 2$) is not merely unproved by [2, Thm. 14]: it is *unprovable by that method*. Let $\eta = e^{i(\omega + \pi/2)}$, so that $f\eta^2 + e\eta + f = 0$ and every edge direction of every such tiling lies in $\langle \eta, -1 \rangle$. Any T -junction-proof invariant of the shape $\sum_{\text{edges}} L \cdot \chi(\theta)$ with $\chi(-w) = -\chi(w)$ — i.e. exactly the Φ /colouring method, and exactly Beeson's — is equivalent to the master identity

$$m(-cz^4 + az^3 + bz^2 + az - c) = z^3(f - ez)P(z) + (fz - e)Q(z), \quad P, Q \in \mathbb{Z}[z, z^{-1}],$$

whose $z = \pm 1$ specializations return exactly $M = m(f + e)$ and $-m(f - e)$. But that identity is *solvable for every* m , with the explicit witness $P_0 = mf(z^{-1} - z^{-3})$, $Q_0 = m(ez^2 - fz^3)$. Hence no invariant in this class can force $g \mid M$, for any (e, f, m) : the Φ -method — the engine of both Beeson's argument and of our own $2\pi/3$ theorem — provably cannot decide the base- β residue. Closing it requires genuinely non-linear input.

Remark 8 (What the $e \geq 2$ residue costs, and where it now sits). The $e \geq 2$ residue ($N = 23, 59, 71, 83, 131, 167, \dots$) resists all three natural methods, and in two cases *provably*:

- **Linear/colouring invariants** — dead by Remark 7.
- **b -edge census**. On a maximal segment of length L the two sides carry $q_1 \equiv q_2 \pmod{f}$ (both $\equiv -e^{-2}L$), and the entire census — in *every* modulus, and in its per-direction

refinement — is a formal consequence of the single parity identity $N(a+b+c) \equiv 2X+Y \pmod{2}$, valid for all (e, f, m) . So no congruence of this type can bite.

- **Euler / vertex counting.** For any such tiling $V - E + F = 2$ is *identically* the sum of the three corner identities $\#\alpha = \#\beta = \#\gamma = N$; the vertex-type system collapses to $Q_{013} = 1 + P_{320} + 2Q_{640} + Q_{431}$ and $3P_{320} + P_{111} + 6Q_{640} + 4Q_{431} + 2Q_{222} = N - 3$, which is feasible for every $e \geq 2, m = 1$ target. So Euler carries no information here.

What *is* gained is citation-free structure, valid for all (e, f, m) and validated against the 44-tiling: (a) γ -trap — no boundary junction carries γ on both sides, hence every side of ABC carries at least one c -edge, and at $m = 1$ *neither equal side carries any b -edge*, so all boundary b -edges lie on the base; (b) *corner parallelogram* — the unique β -corner tile T_B has its b -edge a chord $[P, Q]$ matched by a single tile T_3 whose α, γ are swapped, so $T_B \cup T_3$ is a parallelogram with sides a, c ; consequently the first two and last two base edges, and the first two edges of each equal side, lie in $\{a, c\}$.

This isolates the residue to one clean statement. Since $q_{\text{base}} \equiv em \pmod{f}$ while $q_{\text{equal}} \equiv 0$, we have:

if the base carries no b -edge, then $f \mid em$, hence $f \mid m$, hence no tiling with $m = 1$ — for every e at once.

The only known member of the family, the 44-tiling ($m = 2$), indeed carries *zero* b -edges on its entire boundary, consistent with this. Proving “the base of a base- β tiling never carries a b -edge” would close the whole gap — including the $e = 1$ cases now resting on [2, Lem. 6] — and by Remark 7 it cannot be proved by any linear edge-direction invariant. We leave it as the sharp open problem.

Remark 9 (The colouring theorem is citation-free). The same identity has a positive consequence: it *proves* the sound half of [2, Thm. 14] rather than citing it. Because α/π is irrational (Remark 7’s η , via Niven — `tile_alpha_irrational`), η is not a root of unity: $n(\frac{\alpha}{2} + \frac{\pi}{2}) = k\pi$ forces $n = 0$ (`direction_free, colouring_well_defined`, both axiom-clean). Hence $\langle \eta, -1 \rangle \cong \mathbb{Z} \times \mathbb{Z}/2$ and the sign character $\chi_2(\pm\eta^n) = \pm(-1)^n$ is *well defined* — which is exactly what makes the colouring number $M = \sum_t \chi_2(d_t)$ a well-defined integer, and the $z = -1$ specialization of the master identity then returns $M = m(f + e)$ together with the size formulas $X = f^3 M / (f + e)$, $Y = eM(3f^2 - e^2) / (f + e)$. Consequently the $3\alpha + 2\beta$ tiling equations — and with them the necessary sides of the triquadratic and four-component branches (Remark 2) — no longer rest on a preprint’s colouring theorem: the direction-group foundation is machine-checked, and the cancellation step is the same Φ -style argument proved in Section 4.

Proposition 10 (Base- $(\alpha+\beta)$ prime exclusion, correcting [2, Thm. 18]). *No isosceles triangle with base angles $\alpha + \beta$, where $3\alpha + 2\beta = \pi$ and $\alpha \notin \mathbb{Q}\pi$, is N -tiled for N prime.*

Proof. Fix a putative tiling with coloring number M ; by [2, Cor. 3], $0 < M^2 < N$. Put $d = \gcd(N - M^2, N + M^2)$. As $d \mid 2N$, $d \mid 2M^2$, and $\gcd(N, M^2) = 1$ (a prime $N > M^2$ cannot divide M^2), $d \mid 2$. Writing $g = (N + M^2)/d$ and $\hat{a} = (N - M^2)/d$, the primitive tile is $(a, b, c) = (g\hat{a}, g^2 - \hat{a}^2, g^2)$ with $0 < \hat{a} < g$ ([2, Thm. 17, Lem. 8]), and the base AC has integer length $Y = M\hat{a}(g + \hat{a})$ ([2, Lem. 42]). A tiling forces the base to carry at least two c -edges ([2, Lem. 6]) and a number of b -edges $\equiv -M \pmod{g}$, hence at least $g - M > 0$ ([2, Lem. 45(iii)]); so $Y \geq (g - M)b + 2c$. But $d \mid 2$ gives $\hat{a} + M \leq g$, and the identity

$$(g - M)(g^2 - \hat{a}^2) - M\hat{a}(g + \hat{a}) = g(g + \hat{a})(g - \hat{a} - M) \geq 0$$

gives $Y \leq (g - M)b < (g - M)b + 2c$, a contradiction. (E.g. $N = 79$, $M = 1$: $d = 2$, $g = 40$, $\hat{a} = 39$, $b = 79$, $c = 1600$, so $Y = 3081$ while $(g - M)b + 2c = 6281$.) The two arithmetic facts used — $d \mid 2$ and the identity together with $\hat{a} + M \leq g$ — are machine-checked in Lean (`lean/BaseAlphaBetaPrime.lean`, axiom-clean); only the covering bounds are cited. $\square$

The tile. Throughout, T has angles (α, β, γ) opposite sides (a, b, c) , with $\gamma = 2\pi/3$, hence $\alpha + \beta = \pi/3$; we treat the incommensurable case, α an irrational multiple of π . Once the target is known isosceles (Section 3), T has commensurable sides by [3, Thm. 12.5] (refereed); we scale to

$$(a, b, c) \in \mathbb{Z}^3, \quad \gcd(a, b, c) = 1, \quad c^2 = a^2 + ab + b^2 \quad (1)$$

(the law of cosines at $\gamma = 2\pi/3$). We call such a triple a *primitive 120°-triple*. If a prime p divides two of a, b, c it divides the third (by (1)), so (1) gives $\gcd(a, b) = \gcd(a, c) = \gcd(b, c) = 1$. One computes

$$\sin \alpha = \frac{a\sqrt{3}}{2c}, \quad \cos \alpha = \frac{2b+a}{2c}, \quad \sin \beta = \frac{b\sqrt{3}}{2c}, \quad \cos \beta = \frac{2a+b}{2c}. \quad (2)$$

The direction grid. Set $G = \{j\frac{\pi}{3} + k\alpha : j, k \in \mathbb{Z}\} \pmod{2\pi}$. We claim every edge of every tile in a tiling points in a direction lying in G , for a single fixed reference. Fix the reference along an edge of one tile; its three edge-directions are then in G , since the tile angles $\alpha, \beta = \frac{\pi}{3} - \alpha, \gamma = \frac{2\pi}{3}$ all lie in $\mathbb{Z}\alpha + \mathbb{Z}\frac{\pi}{3}$. Now argue by adjacency. The tiling fills a connected region, so any tile t' shares at least a boundary point P with the union of the tiles already placed; at P the surrounding corner angles each belong to $\{\alpha, \beta, \gamma\}$, and a directed edge of t' at P differs from a known grid direction at P by a sum of such angles, hence lies in G ; the remaining edges of t' then differ from it by $\mathbb{Z}$ -combinations of $\alpha, \frac{\pi}{3}$, so all of t' is in G . This uses only the angles meeting at a point and so applies verbatim at non-edge-to-edge incidences, where P lies in the relative interior of an edge (the angles there sum to π rather than 2π). Since α/π is irrational, $\frac{\pi}{3}$ and α are $\mathbb{Q}$ -linearly independent, so the pair (j, k) is determined by the direction (and $(-1)^j$ by the direction alone, as $2\pi = 6 \cdot \frac{\pi}{3}$).

Vertex types and shapes of ABC . A corner of ABC is filled by tile-corners: if $P, Q, R \geq 0$ (not all 0) of them are α -, β -, γ -corners, the corner measures $P\alpha + Q\beta + R\gamma = (P - Q)\alpha + (Q + 2R)\frac{\pi}{3}$. Write $(m, k) := (P - Q, Q + 2R)$. Since $Q \equiv k \pmod{2}$, $0 \leq Q \leq k$, and $P = m + Q \geq 0$, the pair (m, k) is *realizable* exactly when

$$k \geq 1 \text{ and } m \geq -k, \quad \text{or} \quad k = 0 \text{ and } m \geq 1. \quad (3)$$

The three corners of ABC have angles $m_i\alpha + k_i\frac{\pi}{3}$ summing to $\pi = 3 \cdot \frac{\pi}{3}$; as α is irrational this forces

$$\sum_i m_i = 0, \quad \sum_i k_i = 3. \quad (4)$$

By (3) each $m_i + k_i \geq 0$, and by (4) $\sum_i (m_i + k_i) = 3$; hence each $m_i + k_i \in \{0, 1, 2, 3\}$ and $m_i \in [-k_i, 3 - k_i] \subseteq [-3, 3]$. The enumeration is therefore finite and *independent of α* . Checking the finitely many triples $\{(m_i, k_i)\}$ satisfying (3), (4) and $0 < m_i\alpha + k_i\frac{\pi}{3} < \pi$ (done in exact arithmetic in Section 8) leaves exactly eleven shapes of ABC : the equilateral triangle; two isosceles triangles, with base angles α (apex $\alpha + 3\beta$) and β (apex $\beta + 3\alpha$); and eight scalene triangles, namely

$$F_1 = (\alpha, \alpha + \beta, \alpha + 2\beta), \quad F_2 = (2\alpha, 2\beta, \alpha + \beta), \quad F_3 = (\alpha, 2\alpha, 3\beta), \quad F_4 = (\alpha, 2\beta, 2\alpha + \beta),$$

their reflections under $\alpha \leftrightarrow \beta$ (with F_2 self-paired), and the tile-similar triangle $(\alpha, \beta, 2\pi/3)$. Since α/π is irrational, distinct pairs (m, k) give distinct angles, so the labels *equilateral*, *isosceles*, *scalene* attached to these types are unambiguous: no accidental coincidence of angles can occur, as any such coincidence would force $\alpha \in \mathbb{Q}\pi$. These types coincide with the six sporadic non-equilateral $2\pi/3$ similarity types of Zhang [7].

The area identity. If ABC is similar to a fixed shape whose primitive integer side-proportion vector is D (gcd of its entries 1), and ABC is N -tiled, then each side of ABC is a disjoint union of whole tile-edges (a tile meeting a straight portion of the boundary has a whole side on it), hence an integer; so the sides of ABC are $k \cdot D$ for an integer $k \geq 1$, and

$$N = \frac{\text{Area}(ABC)}{\text{Area}(T)} = k^2 N_0, \quad N_0 := \frac{\text{Area}(D)}{\text{Area}(T)}, \quad \text{Area}(T) = \frac{\sqrt{3}}{4} ab. \quad (5)$$

This is a necessary condition on N .

3. REDUCTION TO THE ISOSCELES CASE

Proposition 11. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. If T N -tiles a triangle and N is prime, then ABC is isosceles and not equilateral.*

Proof. By Section 2, the corner-type triple of ABC is the equilateral one, the tile-similar one, one of $F_1, \dots, F_4$ or their reflections, or one of the two isosceles ones. The equilateral case gives N not prime [5, Thm. 6]; the tile-similar case gives $N = n^2$ [1, 12], not prime. For the remaining non-isosceles shapes we compute N_0 from (5); in each case $N = k^2 N_0$ is composite.

F_2 and F_4 . The side-proportion vector of F_2 is $(a(a+2b), b(2a+b), c^2)$ with the $\pi/3$ vertex between the first two sides, giving $N_0 = (a+2b)(2a+b)$. For F_4 one checks the side-vector $(ac, b(2a+b), (a+b)c)$ has gcd 1 (any common prime g divides c and then $g^2 \mid 3b^2$, forcing $g = 1$), giving $N_0 = (a+b)(2a+b)$. Both are products of two integer factors each at least 3, hence composite, so $N = k^2 N_0$ is composite.

F_3 . The law of sines gives the side-proportion vector $(ac^2, a(a+2b)c, 3ab(a+b))$, whose entries share the factor a ; dividing it out leaves $(c^2, (a+2b)c, 3b(a+b))$, and $N_0 = 3(a+b)(a+2b)$. This is divisible by 3, hence composite, so $N \equiv 0 \pmod{3}$ is composite.

F_1 . The side-vector is $(a, c, a+b)$ with the $\pi/3$ vertex between the sides a and $a+b$, so $N_0 = (a+b)/b$ and $N = k^2(a+b)/b$. As $\gcd(b, a+b) = 1$, integrality forces $b \mid k^2$, so $N = t(a+b)$ with $t = k^2/b \in \mathbb{Z}_{\geq 1}$. It remains to show $a+b$ is composite. Suppose $a+b = p$ is prime. Since $(a+b)^2 - c^2 = ab$ by (1), we have $ab = p^2 - c^2$, so a, b are the roots of $x^2 - px + (p^2 - c^2)$, whose discriminant $4c^2 - 3p^2$ must be a perfect square d^2 ; then $3p^2 = (2c-d)(2c+d)$. With $2c-d < 2c+d$ the factorizations of $3p^2$ into two positive factors are $(1, 3p^2)$, $(3, p^2)$, $(p, 3p)$, and in each the smaller root $\frac{p-d}{2}$ is not a positive integer: in the first, $d = (3p^2 - 1)/2 > p$, so the root is negative; in the second, $d = (p^2 - 3)/2$ and the root $\frac{p-d}{2} = -\frac{(p-3)(p+1)}{4}$ is negative for $p > 3$; in the third, $d = p$ gives $c = p$ and $ab = p^2 - c^2 = 0$. (The case $p \leq 3$ is impossible since $a+b \geq 8$ for a primitive 120° -triple.) Hence $a+b \geq 8$ is composite, and $N = t(a+b)$ is composite.

The reflected shapes are handled by $\alpha \leftrightarrow \beta$ (which swaps $a \leftrightarrow b$ and leaves each N_0 above of the same form). So a prime N forces the corner-type triple of ABC to be one of the two isosceles types; those have angles $(\varphi, \varphi, \pi - 2\varphi)$ with $\varphi \in \{\alpha, \beta\}$, and they are never equilateral ($\varphi = \frac{\pi}{3}$ would force $\beta = 0$ or $\alpha = 0$). Hence ABC is isosceles and not equilateral. $\square$

4. THE SIGNED-DIRECTION INVARIANT

Fix the reference so the direction grid is $G = \{j\frac{\pi}{3} + k\alpha\}$. Because $\beta = \frac{\pi}{3} - \alpha$, every grid direction also has the form $(j+k)\frac{\pi}{3} - k\beta$, so we may read off the coefficient of $\frac{\pi}{3}$ in two coordinate systems. Define, for a direction $\theta \in G$,

$$f_\alpha(\theta) = (-1)^j \quad (\theta = j\frac{\pi}{3} + k\alpha), \quad f_\beta(\theta) = (-1)^{j'} \quad (\theta = j'\frac{\pi}{3} + k'\beta).$$

Both are well defined (Section 2), and both satisfy

$$f(\theta + \pi) = -f(\theta), \quad (6)$$

since adding $\pi = 3 \cdot \frac{\pi}{3}$ increases the $\frac{\pi}{3}$ -coefficient by 3. They are genuinely different functions: $f_\beta = (-1)^k f_\alpha$.

For $f \in \{f_\alpha, f_\beta\}$ and a directed edge of length L and direction θ , assign the weight $L f(\theta)$. For a tile t with boundary traversed counterclockwise, put $C_f(t) = \sum_{\text{edges}} L f(\theta)$, and for the boundary of ABC put $\Phi_f(\partial ABC) = \sum_{\text{sides}} L f(\theta)$ (counterclockwise).

Lemma 12 (Cancellation). *For each $f \in \{f_\alpha, f_\beta\}$, $\sum_{\text{tiles}} C_f(t) = \Phi_f(\partial ABC)$.*

Proof. For a grid direction θ let $\Lambda(\theta)$ be the total length of all directed tile-edges of direction θ , summed over every tile (counterclockwise) and every edge; then, grouping the double sum $\sum_t C_f(t)$ by direction,

$$\sum_{\text{tiles}} C_f(t) = \sum_{\theta} \Lambda(\theta) f(\theta). \quad (7)$$

Split each tile-edge into its maximal sub-segments that are either interior to ABC or contained in ∂ABC ; this only refines the sums and changes nothing. Consider the finite set of maximal straight segments S carved out on the tiling's edges. If S lies in the interior of ABC , a neighbourhood of a generic point of S is a disc split by S into two half-discs, each tiled; the tiles on one side meet S in directed edges of total length $|S|$ pointing one way, say θ , and those on the other side in directed edges of total length $|S|$ pointing $\theta + \pi$ (the two subdivisions may differ, a non-edge-to-edge incidence, but each side covers S exactly once because the tiles cover ABC with disjoint interiors). If instead $S \subseteq \partial ABC$, only the single tile inside ABC meets it, contributing length $|S|$ in the counterclockwise direction of ∂ABC . Writing $\Lambda = \Lambda_{\text{int}} + \Lambda_{\text{bd}}$ accordingly, interior segments give $\Lambda_{\text{int}}(\theta) = \Lambda_{\text{int}}(\theta + \pi)$ for every θ , while $\Lambda_{\text{bd}}(\theta)$ is the counterclockwise boundary length in direction θ . Substituting into (7) and pairing θ with $\theta + \pi$,

$$\sum_{\{\theta, \theta+\pi\}} \Lambda_{\text{int}}(\theta) (f(\theta) + f(\theta + \pi)) = 0$$

by (6), so $\sum_{\text{tiles}} C_f(t) = \sum_{\theta} \Lambda_{\text{bd}}(\theta) f(\theta) = \Phi_f(\partial ABC)$. $\square$

Lemma 13 (Tile value). *For every placement of T , $C_{f_\alpha}(t) = \pm(c + a - b)$ and $C_{f_\beta}(t) = \pm(c + b - a)$.*

Proof. Take a positively oriented placement and label the vertices V_1, V_2, V_3 in counterclockwise order with interior angles α, β, γ respectively; let e_i be the directed edge from V_i to V_{i+1} (indices mod 3). Then e_1, e_2, e_3 are opposite V_3, V_1, V_2 , so they have lengths c, a, b . Traversing counterclockwise turns left at V_{i+1} by the exterior angle $\pi - (\text{angle at } V_{i+1})$, so $\text{dir}(e_2) = \text{dir}(e_1) + (\pi - \beta)$ and $\text{dir}(e_3) = \text{dir}(e_2) + (\pi - \gamma)$. Now π adds 3 to the $\frac{\pi}{3}$ -coefficient j , while α adds 0, $\beta = \frac{\pi}{3} - \alpha$ adds 1, and $\gamma = \frac{2\pi}{3}$ adds 2; so $\pi - \beta$ adds $3 - 1 = 2$ and $\pi - \gamma$ adds $3 - 2 = 1$. Writing j_0 for the $\frac{\pi}{3}$ -coefficient of $\text{dir}(e_1)$, the three coefficients are $j_0, j_0 + 2, j_0 + 3$, so $f_\alpha = (-1)^{j_0}$ on e_1, e_2 and $-(-1)^{j_0}$ on e_3 . As e_1, e_2, e_3 carry lengths c, a, b ,

$$C_{f_\alpha}(t) = (-1)^{j_0} (c + a - b).$$

Any placement is obtained from this one by a grid rotation $m\alpha + n\frac{\pi}{3}$, which shifts every j by n and multiplies C_{f_α} by $(-1)^n$, possibly followed by a reflection, which reverses the traversal and negates C_{f_α} ; hence $C_{f_\alpha}(t) = \pm(c + a - b)$ in every case. Repeating in the β -frame (where $\gamma = \frac{2\pi}{3}$ adds 0 to j' and α adds 1) makes the a -edge the odd one out, giving $C_{f_\beta}(t) = \pm(c + b - a)$. $\square$

Corollary 14 (Integrality and parity). *Each of $M_\alpha = \frac{\Phi_{f_\alpha}(\partial ABC)}{c + a - b}$ and $M_\beta = \frac{\Phi_{f_\beta}(\partial ABC)}{c + b - a}$ is an integer, and moreover*

$$M_\alpha \equiv M_\beta \equiv N \pmod{2}.$$

They are separate necessary conditions: each follows from Lemma 12 for its own f , so a single failure forbids a tiling.

Proof. By Lemmas 12 and 13, $\Phi_{f_\alpha}(\partial ABC) = \sum_t C_{f_\alpha}(t) = (c + a - b) \sum_t \varepsilon_t$ with $\varepsilon_t \in \{\pm 1\}$, so $M_\alpha = \sum_t \varepsilon_t$ is a sum of N signs, whence $M_\alpha \equiv N \pmod{2}$; likewise for f_β . $\square$

5. THE ISOSCELES OBSTRUCTION

Lemma 15 (Non-integrality). *Let a, k be positive integers with $\gcd(a, k) = 1$, put $b = k^2$, and let $c > 0$ satisfy $c^2 = a^2 + ab + b^2$. Then $k \nmid (a + b - c)$.*

Proof. From $c^2 = a^2 + ab + b^2$ we get $c > a$ and, since $c^2 < (a + b)^2$, also $c < a + b$. Suppose $k \mid (a + b - c)$. As $b = k^2 \equiv 0 \pmod{k}$, this gives $c \equiv a \pmod{k}$; with $0 < c - a < b = k^2$ we may write $c = a + kt$, $1 \leq t \leq k - 1$. Substituting into $c^2 = a^2 + ak^2 + k^4$ and dividing by k ,

$$2at + kt^2 = ak + k^3, \quad \text{that is} \quad a(2t - k) = k(k - t)(k + t).$$

Then $k \mid a(2t - k)$, and $\gcd(a, k) = 1$ gives $k \mid (2t - k)$, hence $k \mid 2t$. But $2 \leq 2t \leq 2k - 2$ contains no multiple of k other than k itself, so $2t = k$. The left side is then $a(2t - k) = 0$, while the right side is $k \cdot \frac{k}{2} \cdot \frac{3k}{2} = \frac{3k^3}{4} > 0$, a contradiction. Hence $k \nmid (a + b - c)$. $\square$

Theorem 16. *Let T be a tile with angles $(\alpha, \beta, 2\pi/3)$, α/π irrational. Then no prime number of copies of T tiles an isosceles, non-equilateral triangle.*

Remark 17. The irrationality hypothesis cannot be dropped: the commensurable tile $(\frac{\pi}{6}, \frac{\pi}{6}, \frac{2\pi}{3})$ tiles an equilateral (hence isosceles) triangle with $N = 3$, by the fan from the centre; that tile lives in the commensurable branch of the classification, where $N = 3n^2$ is allowed. The equilateral target for an *incommensurable* $2\pi/3$ tile is excluded for primes by Beeson [5], and is covered in the assembly of Section 6; it is not part of this theorem.

Proof. ABC has exactly two equal corner angles. Each corner angle is $m\alpha + k\frac{\pi}{3}$ (Section 2); since α/π is irrational, two corner angles are equal only if their pairs (m, k) coincide. So the corner-type triple of ABC has exactly one repeated type, and among the eleven triples of Section 2 exactly the two isosceles ones qualify. Hence the base angle is α or β . We use the invariant generated by the base angle, so that the equal sides land on $\frac{\pi}{3}$ -coefficient 3 (odd) and pick up $f = -1$.

If the base angle is α , then $ABC = k(c, c, 2b + a)$ (equal sides kc , base $k(2b + a)$): the base is at direction 0 ($f_\alpha = +1$) and the equal sides at $\pi - \alpha = 3\frac{\pi}{3} - \alpha$ ($f_\alpha = -1$), so $\Phi_{f_\alpha}(\partial ABC) = k(2b + a) - 2kc = k(2b + a - 2c)$. The tiling count is $N = k^2(a + 2b)/b$; since N is prime and $\gcd(a + 2b, b) = 1$, we have $k^2 = b$, i.e. $k = \sqrt{b}$. By Corollary 14 and $c^2 = a^2 + ab + b^2$,

$$\frac{k(2b + a - 2c)}{c + a - b} = \frac{c - a - b}{k} \in \mathbb{Z}, \quad k = \sqrt{b},$$

so $k \mid (a + b - c)$, contradicting Lemma 15. If the base angle is β , then $ABC = k(c, c, 2a + b)$, the equal sides are at $\pi - \beta$, and using f_β (for which the equal sides have $\frac{\pi}{3}$ -coefficient 3) the same computation gives, with $k = \sqrt{a}$, $(c - a - b)/k \in \mathbb{Z}$, again contradicting Lemma 15 (the integer $a + b - c$ is symmetric in a, b). In both cases no tiling exists. $\square$

Remark 18. The role of the two invariants is essential. For the base-angle- β shape, f_α places the equal sides at $\pi - \beta = \frac{2\pi}{3} + \alpha$, an *even* $\frac{\pi}{3}$ -coefficient, so f_α gives an integer value and does not obstruct; f_β does. Each invariant is a necessary condition in its own right (Corollary 14), so using the one that fails is legitimate, not a choice of convenient frame: on any genuine tiling both invariants return an integer.

6. RESOLUTION OF THE PROBLEM

Proof of Theorem 1. Let $p > 3$ be prime with $p \equiv 3 \pmod{4}$, and suppose some triangle is p -tiled. By the classification (Section 2) the tile is in one of the listed types, and every type other than a $2\pi/3$ tile on a non-equilateral triangle excludes such a p by an existing theorem [12, 1, 4, 2, 3, 5]. For a $2\pi/3$ tile, Proposition 11 forces ABC isosceles and not equilateral, and Theorem 16 excludes that. Hence no such tiling exists. Finally $19 > 3$ is prime and $\equiv 3 \pmod{4}$, so no triangle is cut into 19 congruent triangles. $\square$

7. TOWARDS THE FULL PROBLEM

Erdős's question asks for *all* N . Theorem 1 settles the negative side for primes; this section records what the present methods contribute to the full determination: a complete answer for primes, and, for each sporadic $2\pi/3$ branch, a determination of its *admissible spectrum* — the set of counts compatible with the area and invariant conditions.

Theorem 19 (The prime case, complete). *A prime p is a number of congruent triangles into which some triangle can be cut if and only if $p = 2$, $p = 3$, or $p \equiv 1 \pmod{4}$.*

Proof. If. For $p = 2$, the altitude from the right angle splits an isosceles right triangle into two congruent triangles. For $p = 3$, join the centre of an equilateral triangle to its vertices. For $p \equiv 1 \pmod{4}$, Fermat's two-squares theorem gives $p = e^2 + f^2$ with $e, f \geq 1$; splitting a right triangle with legs e, f by the altitude from the right angle and tiling the two similar pieces quadratically with e^2 and f^2 congruent copies realizes $N = e^2 + f^2$ (the biquadratic tiling [13, 3]). *Only if.* Theorem 1, whose dependencies this direction inherits. $\square$

Theorem 20 (Isosceles admissible spectrum). *Let (a, b, c) be a primitive 120° -triple and write $b = de^2$ with d squarefree. Every N -tiling of the base- α isosceles target has scale $k = dew$ for a positive integer w , and*

$$N = dw^2(a + 2b), \quad e \mid w(c - a - b), \quad \frac{w(c - a - b)}{e} \equiv N \pmod{2}.$$

Symmetrically, with $a = d'e'^2$, the base- β target forces $N = d'w^2(2a + b)$, $e' \mid w(c - a - b)$ and the same parity for $w(c - a - b)/e'$.

Proof. By the area identity $Nb = k^2(a + 2b)$ and $\gcd(a + 2b, b) = 1$ we get $b \mid k^2$. For a prime $p \mid d$ the valuation $v_p(b) = 2v_p(e) + 1 \leq 2v_p(k)$ forces $v_p(k) \geq v_p(e) + 1$, and for $p \nmid d$, $v_p(k) \geq v_p(e)$; hence $de \mid k$, say $k = dew$, and substituting back, $N = dw^2(a + 2b)$. By Corollary 14 and the identity $(c - a - b)(c + a - b) = b(2b + a - 2c)$, the invariant value is $M = k(c - a - b)/b$, so $b \mid k(c - a - b)$, i.e. $de^2 \mid dew(c - a - b)$, i.e. $e \mid w(c - a - b)$; and $M = w(c - a - b)/e \equiv N \pmod{2}$ by Corollary 14. $\square$

Theorem 16 is the degenerate case: N prime forces $d = w = 1$, so $b = e^2$ and $e \mid (c - a - b)$, which Lemma 15 refutes. The residual divisibility is governed by the following classification.

Proposition 21 (Solvability of $e \mid (a + b - c)$). *Let $b = de^2$, d squarefree, $\gcd(a, b) = 1$. Then $e \mid (a + b - c)$ holds for a primitive 120° -triple if and only if there is an integer j with $d < j < 2d$ such that ej is even,*

$$a = \frac{e^2(2d - j)(2d + j)}{4(j - d)}$$

is a positive integer coprime to de^2 ; in that case $c = a + \frac{e^2 j}{2}$. In particular, for $d = 1$ the range $(1, 2)$ contains no integer, which re-proves Lemma 15; for $d = 2$, $j = 3$, $e = 2$ one obtains $(a, b, c) = (7, 8, 13)$.

Proof. Since $e \mid b$, the hypothesis gives $c \equiv a \pmod{e}$, and $0 < c - a < b$ writes $c = a + et$ with $1 \leq t \leq de - 1$. Substituting into $c^2 = a^2 + ab + b^2$ and dividing by e , $a(2t - de) = e(de - t)(de + t)$. As $e \mid a(2t - de)$ and $\gcd(a, e) = 1$, we get $e \mid 2t$, say $2t = ej$ with $1 \leq j \leq 2d - 1$; substituting $t = ej/2$ gives $4a(j - d) = e^2(2d - j)(2d + j)$, whose right side is positive, forcing $j > d$ and the displayed formula. Conversely, given such j , setting $t = ej/2$, a as displayed and $c = a + et$ reverses the computation: $c^2 = a^2 + ab + b^2$ and $a + b - c = e(de - t)$ is divisible by e . $\square$

Theorem 22 (Spectrum lattice). *Let (a, b, c) be a primitive 120° -triple, $b = de^2$ with d square-free, $r = c - a - b$, $g = \gcd(e, r)$, $e_1 = e/g$, and*

$$T = \frac{r}{g} + de_1^2(a + 2b) \pmod{2}.$$

Set $E = e_1$ if T is even and $E = 2e_1$ if T is odd. Then the counts admissible for the base- α isosceles target (Theorem 20 with the parity clause) are exactly

$$N = d(Eu)^2(a + 2b), \quad u = 1, 2, 3, \dots$$

Proof. By Theorem 20 the conditions on w are $e \mid wr$ and $wr/e \equiv N \pmod{2}$. As $\gcd(e_1, r/g) = 1$, the divisibility is equivalent to $e_1 \mid w$; writing $w = e_1u$, the count is $M = ur/g$, and using $u^2 \equiv u \pmod{2}$,

$$M - N = \frac{ur}{g} - de_1^2u^2(a + 2b) \equiv u\left(\frac{r}{g} + de_1^2(a + 2b)\right) = uT \pmod{2},$$

so the parity clause holds for all u when T is even, and exactly for even u when T is odd. $\square$

Theorem 22 uses only the base-aligned invariant. The mirror invariant is also a necessary condition on the same target, and it closes the spectrum completely.

Theorem 23 (Spectrum theorem). *Let (a, b, c) be a primitive 120° -triple. The tile counts satisfying all the invariant conditions on the base- α isosceles target — integrality and parity of both M_α and M_β , together with the area equation — are exactly Zhang's family*

$$N = m^2b(a + 2b), \quad m = 1, 2, 3, \dots$$

Symmetrically, the base- β target admits exactly $N = m^2a(2a + b)$, and the F_1 target exactly $N = m^2b(a + b)$.

Proof. Write $X = c + a - b$, $Y = c + b - a$, so $XY = 3ab$ by (1). On the base- α target with scale k , $M_\alpha = k(2b + a - 2c)/X$ and $M_\beta = k(2b + a + 2c)/Y$ (the equal sides carry $\frac{\pi}{3}$ -coefficient 3 in the α -frame and 2 in the β -frame). Since $(2b + a - 2c) + 2X = 3a$ and $(2b + a + 2c) - 2Y = 3a$, the two integralities are equivalent to

$$X \mid 3ak \quad \text{and} \quad Y \mid 3ak,$$

and, setting $A = 3ak/X$, $B = 3ak/Y$, the parity clauses of Corollary 14 read $A \equiv B \equiv N \pmod{2}$. Recall $k = dew$ and $N = dw^2(a + 2b)$, with $b = de^2$.

If $e \mid w$, say $w = em$, then $k = de^2m = bm$, so $A = Ym$ and $B = Xm$ are integers, and $A - N \equiv m(Y + b(a + 2b)) \pmod{2}$ using $m^2 \equiv m$; now $Y + b(a + 2b) \equiv c + a + b + ab \equiv 0 \pmod{2}$, because (1) gives $c \equiv a + b + ab$; likewise for B . So every member of Zhang's family is admissible.

Conversely, fix a prime p and let $x, y, \sigma, \varepsilon, \delta, \omega$ be the p -adic valuations of $X, Y, 3a, e, d, w$; then $x + y = \sigma + \delta + 2\varepsilon$ and the divisibilities give $\max(x, y) \leq \sigma + \delta + \varepsilon + \omega$, so $\omega \geq \varepsilon - \min(x, y)$. A common prime of X and Y divides $X + Y = 2c$ and $Y - X = 2(b - a)$; for odd p this forces $p \mid c$ and $p \mid b - a$, whence $3 \mid c$ or $p \mid \gcd(a, b)$ — both impossible ($3 \nmid c$ as in Proposition 11). So $\min(x, y) = 0$ for every odd p , and $\omega \geq \varepsilon$ there. At $p = 2$: if ab is odd then X, Y are odd and the same argument applies. If b is even (so a is odd), a two-line computation modulo 8 shows

$v_2(b) \notin \{1, 2\}$: $b \equiv 2 \pmod{4}$ gives $c^2 \equiv 1 + 2 \equiv 3 \pmod{4}$, and $v_2(b) = 2$ gives $c^2 \equiv 1 + 4 \equiv 5 \pmod{8}$, both impossible. So $v_2(b) \geq 3$. Here $v_2(X + Y) = v_2(2c) = 1$ forces $\min(x, y) = 1$, and the divisibilities alone allow $\omega = \varepsilon - 1$; but then exactly one of A, B is odd (their 2-valuations are 0 and $v_2(b) - 2$), while $v_2(N) = \delta_2 + 2\omega = v_2(b) - 2 \geq 1$ makes N even — so the parity clause fails, and $\omega \geq \varepsilon$ at $p = 2$ as well. Hence $e \mid w$, i.e. $N = m^2 b(a + 2b)$. The base- β case is the mirror $a \leftrightarrow b$. For F_1 the counts are $M_\alpha = 3ak/X - k$ and $M_\beta = 3ak/Y + k$ (Section 7), so the same divisibilities appear, the parity clauses read $A \equiv B \equiv N + k \pmod{2}$, and in the boundary case $v_2(k) = \delta_2 + 2\varepsilon_2 - 1 \geq 2$ makes $N + k$ even against an odd count; the argument is otherwise identical, giving $N = m^2 b(a + b)$. $\square$

Remark 24. The single-invariant lattice of Theorem 22 can be strictly coarser than $e\mathbb{Z}$ (the smallest example is $(11, 24, 31)$, whose lattice admits $N = 354$), which briefly suggested admissible values beyond Zhang’s constructions. Theorem 23 corrects this: the mirror invariant eliminates every such value (354 by parity, 15689 on $(19, 261, 271)$ already by integrality), and the fully refined spectrum *coincides with Zhang’s family on every tile* — verified exhaustively over all 138 primitive triples with $a, b < 400$. The necessary side of the isosceles and F_1 branches is therefore closed. What remains open in these branches is exactly the realizability of the *small* family members. Not all of them are open: a family member that is also a sum of two squares (or 2, 3, 6 times a square) is realized by the commensurable construction [4], so the genuinely undecided members are precisely those with a prime $\equiv 3 \pmod{4}$ to an odd power. (Thus $N = 40 = b(a + b)$ on $(3, 5, 7)$ is realized as $2^2 + 6^2$, and $N = 65 = b(a + 2b)$ on $(3, 5, 7)$ as $1^2 + 8^2$, while $105 = 3 \cdot 5 \cdot 7$ is not a sum of two squares.) Zhang’s constructions cover m in certain residue classes from some bound onward, so the sharpest open instances are the smallest such members: $N = 105 = b(a + b) = 7 \cdot 15$ on the F_1 target $(105, 56, 91)$ of the tile $(8, 7, 13)$, then 120, 132, 154, 184. Each has membership undecided by any known necessary condition, and is exactly the kind of bounded existence question the engine of Section 7 addresses — though for N of this size a complete exhaustive search is not yet computationally in reach.

Remark 25. Admissibility is necessary, not sufficient. The smallest admissible value is $N = 33 = 3 \cdot 11$ on the tile $(5, 3, 7)$ (where $d = 3, e = 1$, so the divisibility is vacuous); it is nevertheless excluded by Beeson’s boundary analysis, which rules out $N < 36$ [3], while Herdt’s 2673-tile tiling is the member $w = 9$ of the same family $33w^2$. The parity clause bites hard: on the tile $(7, 8, 13)$ (where $d = 2, e = 2, c - a - b = -2$) the divisibility admits all $N = 2w^2 \cdot 23$, but the count $M = -w$ must have the parity of $N = 2w^2 \cdot 23$, which is even — forcing w even, say $w = 2m$; the refined spectrum is exactly $N = 184m^2 = m^2 b(a + 2b)$ — *precisely* Zhang’s constructed family [7] for this tile. In particular $N = 46$, previously the smallest value passing every known necessary condition on this branch, is *excluded* (Section 7 checks the remaining branches), in exact agreement with the prediction of Zhang’s conjecture. Zhang’s constructions realize $N = m^2 b(a + 2b)$ for m in certain residue classes; the refined spectrum bounds the realized set from outside, and Theorem 22 determines it exactly.

Proposition 26 (The other branches). *With the same normalizations: F_1 forces $N = dw^2(a + b)$ where $b = de^2$, d squarefree (from $b \mid k^2$ as above); F_2, F_3, F_4 force $N = N_0 k^2$ exactly, with $N_0 = (a + 2b)(2a + b), 3(a + b)(a + 2b), (a + b)(2a + b)$; the tile-similar shape forces $N = n^2$; and an equilateral target of side S forces $ab \mid S^2$, $N = S^2/(ab)$ (so N and ab have the same squarefree kernel) together with $(c + a - b) \mid 3S$ and $(c + b - a) \mid 3S$.*

Proof. The scalene statements are the area identity (5) with the side-vectors of Section 3 (F_1 repeats the $b \mid k^2$ analysis). For the equilateral target the three sides, traversed counterclockwise, differ in direction by $\frac{2\pi}{3} = 2 \cdot \frac{\pi}{3}$, so f takes one sign on all three and $\Phi_f(\partial ABC) = \pm 3S$ for both invariants; Corollary 14 gives the two divisibilities, and the area identity gives $N ab = S^2$. $\square$

The equilateral $2\pi/3$ spectrum. The equilateral divisibilities of Proposition 26 reduce to a clean finite criterion.

Proposition 27 (Equilateral admissibility). *Let an equilateral triangle of side S be N -tiled by a $2\pi/3$ tile (a, b, c) , and put $X = c + a - b$, $Y = c + b - a$. Then $XY = 3ab$, and $s := 3S/X$, $t := 3S/Y$ are positive integers with*

$$st = 3N, \quad s \equiv t \equiv N \pmod{2}, \quad (t - s)^2 + 16N = q^2 \text{ for an integer } q;$$

conversely (s, t, q) determine the instance: (a, b, c) is proportional to $(q + (t - s), q - (t - s), 2(s + t))$, and primitivity pins the scale, hence S .

Proof. $XY = c^2 - (a - b)^2 = 3ab$ by (1). By Proposition 26, $X \mid 3S$ and $Y \mid 3S$, so $s, t \in \mathbb{Z}_{\geq 1}$; indeed $s = M_\alpha$ and $t = M_\beta$ are the two invariant counts, so $s \equiv t \equiv N \pmod{2}$ by Corollary 14, and $st = 9S^2/(XY) = 9Nab/(3ab) = 3N$. From $X - Y = 2(a - b)$ we get $a - b = \frac{3S}{2}(\frac{1}{s} - \frac{1}{t}) = \frac{S(t-s)}{2N}$, and $4ab = 4S^2/N$ gives $(a+b)^2 = (a-b)^2 + 4ab = \frac{S^2}{4N^2}[(t-s)^2 + 16N]$; as $a+b$ is an integer, $q := 2N(a+b)/S$ is a rational with integer square, hence an integer, and $(t-s)^2 + 16N = q^2$. Then $a - b$, $a + b$ and $c = (X + Y)/2 = \frac{S(s+t)}{2N}$ give the displayed proportions. $\square$

The equilateral criteria as divisor conditions. Both equilateral square criteria — ours and Beeson's — complete the square.

Proposition 28 (Conic form). *For an equilateral target: a $2\pi/3$ tile requires a factorization $uv = 16N^2$ with $u + v \equiv 0 \pmod{2}$ and $\frac{u+v}{2} - 5N = s^2$ for a divisor s of $3N$ with $s \equiv 3N/s \equiv N \pmod{2}$; a $\pi/3$ tile requires a factorization $uv = 16N^2$ with $\frac{u+v}{2} = 5N - M^2$ for some $0 < M < \sqrt{N}$, $M \equiv N \pmod{2}$. These are equivalent to the criteria of Proposition 27 and of [5, Thm. 3] respectively.*

Proof. For the $2\pi/3$ case, with $t = 3N/s$ the condition $(t - s)^2 + 16N = q^2$ becomes, after multiplying by s^2 , $(qs)^2 = (3N - s^2)^2 + 16Ns^2 = (s^2 + N)(s^2 + 9N)$, i.e. $(s^2 + 5N)^2 - (qs)^2 = 16N^2$; factor the difference of squares. For the $\pi/3$ case, $(9N - M^2)(N - M^2) = y^2$ is $(5N - M^2)^2 - 16N^2 = y^2$, i.e. $(5N - M^2 - y)(5N - M^2 + y) = 16N^2$. $\square$

The necessary side of the equilateral branch is thus elementary — a divisor condition on $16N^2$ — and, per N , pins finitely many fully determined instances. Both square-completions of Proposition 28 are formalized and machine-checked in Lean (`lean/EquilateralConic.lean`): the elimination of t giving $(qs)^2 = (s^2 + N)(s^2 + 9N)$, the resulting factorization $(s^2 + 5N - qs)(s^2 + 5N + qs) = 16N^2$, and the $\pi/3$ identity $(5N - M^2)^2 - 16N^2 = (9N - M^2)(N - M^2)$, all over $\mathbb{Z}$. The elliptic curves of Laczkovich [11] concern the *realizability* of such instances, not their enumeration.

The values 14 and 15. The machinery above, combined with the published branch theorems, decides every branch of the classification for $N = 14$ and $N = 15$ by finite computations (`verify_frontier.py`): the commensurable, tile-similar, right-angle and $\gamma = 2\alpha$ branches exclude both values by their known N -forms; the sporadic $2\pi/3$ shapes exclude both by the admissible spectra of Theorems 20 and Proposition 26; the $\pi/3$ -tile equilateral case excludes both by Beeson's square criterion ($(9N - M^2)(N - M^2)$ is a perfect square for some $M < \sqrt{N}$ [5, Thm. 3], which fails for every admissible M); and of the five target shapes of the $3\alpha + 2\beta = \pi$ branch [2], four exclude both values through their published tiling equations (checked by exact rational-root enumeration), while the isosceles target with base angles $\alpha + \beta$ admits, for $N = 14$, the single candidate tile $(45, 56, 81)$ with $M = 2$ — which is destroyed by Beeson's boundary congruence $M \equiv -m \pmod{\gcd(a, c)}$ (it forces at least seven b -edges of length 56 on a base of length 140).

What survives everything is four fully explicit finite instances:

N	branch	tile	target
14	equilateral, $2\pi/3$	(7, 8, 13)	equilateral, $S = 28$
15	equilateral, $2\pi/3$	(3, 5, 7)	equilateral, $S = 15$
15	$3\alpha + 2\beta = \pi$, iso- $(\alpha + \beta)$, $M = 1$	(56, 15, 64)	(120, 120, 105)
15	$3\alpha + 2\beta = \pi$, iso- $(\alpha + \beta)$, $M = 3$	(4, 15, 16)	(60, 60, 15)

In each, the tile is *uniquely determined* (Proposition 27 for the equilateral rows; the tiling equation $N/M^2 = (1 + s)/(1 - s)$ of [2, Thm. 17] for the isosceles rows), so each is a single bounded existence question: can 14 (resp. 15) congruent copies of the stated tile fill the stated triangle?

Each instance was settled by an exhaustive, exact search (`code/engine/`). The engine is an advancing-front algorithm over $\mathbb{Q}(\sqrt{D})$ ($D = 3, 23, 7$; every geometric decision exact, no floating point) whose branching rule is provably complete: at the lexicographically least vertex v of the unfilled region — necessarily convex — every tile of any tiling whose closure contains v has a corner there (an edge through v would subtend angle π), so the clockwise-most of them has a corner at v with one side along the clockwise boundary ray; this leaves at most six placements (three corner types, two adjacent sides each, the two choices being mirror images). An exhausted search is therefore a proof of non-existence. Two prunes are used, each a theorem about completable states: the area of every region component is a positive integer multiple of the tile area; and a maximal straight boundary run with *convex* endpoints has length in $\mathbb{N}a + \mathbb{N}b + \mathbb{N}c$ (whole tile edges cannot pass a convex corner — runs with a reflex endpoint are not pruned, as an edge may continue past a reflex corner into the region). The engine passes positive controls — it finds the $N = 4$ reptile subdivisions in both tile families and a hand-built non-edge-to-edge two-tile configuration with T-junctions, each found tiling re-verified by an independent checker — and, as a robustness check, all four verdicts below are unchanged when the run prune is disabled entirely (node counts without it in parentheses).

instance	tile	target	verdict	nodes
$N = 14$, equilateral	(7, 8, 13)	$S = 28$	exhausted, no tiling	133 (482)
$N = 15$, equilateral	(3, 5, 7)	$S = 15$	exhausted, no tiling	156 (604)
$N = 15$, iso- $(\alpha + \beta)$	(56, 15, 64)	(120, 120, 105)	exhausted, no tiling	37 (526)
$N = 15$, iso- $(\alpha + \beta)$	(4, 15, 16)	(60, 60, 15)	exhausted, no tiling	2 (10)

Theorem 29. *No triangle can be cut into 14 congruent triangles, and none into 15.*

Proof. By the branch sweep above, any 14- or 15-tiling would live in one of the four listed instances, each with its tile and target uniquely determined; the exhaustive searches refute all four. The dependencies are those of Theorem 1 together with [2, Thms. 3, 7, 11, 14, 17, 19], [5, Thm. 3], and the declared computer assistance (exact arithmetic, complete branching, independently validated). $\square$

The values 21, 22, 30, and the instance 46. The same sweep, sharpened by the parity clause of Corollary 14, extends further (`verify_frontier.py`). Three remarks carry the new weight. First, the invariant applies to the *scalene* F_1 target: its sides $k(a+b)$, ka , kc have $\frac{\pi}{3}$ -coefficients 0, 2, 3 (the same turning computation as Lemma 13), so

$$M_\alpha = \frac{k(2a + b - c)}{c + a - b}, \quad M_\beta = \frac{k(2a + b + c)}{c + b - a}$$

must be integers of the parity of N . For $N = 21$ this destroys the unique structural candidate, the genuine tile (5, 16, 19) with $k = 4$: $M_\alpha = 28/8 \notin \mathbb{Z}$ — while Zhang’s constructed F_1 family $N = m^2b(a + b)$ passes, so the invariant separates the candidate from the constructions. (For

$N = 30$ the candidate $(7, 8, 13)$, $k = 4$ has $M_\alpha = 3$, killed by parity.) Second, for $N = 22$ and $N = 30$ *every* branch dies arithmetically — the two isosceles- $(\alpha+\beta)$ candidates of 22 fall to the boundary congruence, those of 30 likewise — so no search is needed at all. Third, for $N = 21$ exactly two instances survive, both isosceles- $(\alpha+\beta)$: $M = 1$, tile $(110, 21, 121)$ on $(231, 231, 210)$, and $M = 3$, tile $(10, 21, 25)$ on $(105, 105, 42)$; both live in $\mathbb{Q}(\sqrt{6})$, and the engine exhausts both:

instance	tile	target	verdict	nodes
$N=21$, iso- $(\alpha+\beta)$, $M=1$	$(110, 21, 121)$	$(231, 231, 210)$	exhausted	107 (4266)
$N=21$, iso- $(\alpha+\beta)$, $M=3$	$(10, 21, 25)$	$(105, 105, 42)$	exhausted	15 (102)

Finally, the parity clause settles the value that Remark 25 identified as the sharpest test of Zhang’s conjecture: $N = 46$ passes the divisibility spectrum on $(7, 8, 13)$ but not the parity, and the sweep excludes every other branch, so 46 is not realizable — exactly as Zhang’s conjecture predicts.

The pipeline is now mechanical, and it continues: $N = 38$ and $N = 42$ die in every branch by arithmetic alone (both squarefree, so the $\gamma = 2\alpha$ branch is closed by [3, Thm. 11.7], and no sporadic, equilateral or $3\alpha + 2\beta$ candidate survives), while 33, 35 and 39 each reduce to two isosceles- $(\alpha+\beta)$ instances (the sporadic- $2\pi/3$ candidate $(5, 3, 7)$ at $N = 33$ is excluded by Beeson’s $N < 36$ boundary analysis [3]); all six searches exhaust:

instance	tile	target	nodes
$N=33$, $M=1$	$(272, 33, 289)$	$(561, 561, 528)$	489
$N=33$, $M=3$	$(28, 33, 49)$	$(231, 231, 132)$	49
$N=35$, $M=1$	$(306, 35, 324)$	$(630, 630, 595)$	523
$N=35$, $M=5$	$(6, 35, 36)$	$(210, 210, 35)$	2
$N=39$, $M=1$	$(380, 39, 400)$	$(780, 780, 741)$	853
$N=39$, $M=3$	$(40, 39, 64)$	$(312, 312, 195)$	67

Theorem 30. *No triangle can be cut into 21, 22, 30, 33, 35, 38, 39, 42, or 46 congruent triangles.*

Proof. The sweeps above; for 21, 33, 35 and 39, additionally the exhaustive searches listed. Dependencies as in Theorem 29. $\square$

$N = 44$ **is realizable.** The value 44, the smallest undetermined after Theorem 30, has three live branches. Two die: the isosceles- $(\alpha+\beta)$ instance (tile $(30, 11, 36)$, target $(132, 132, 110)$, from Theorem 17 of [2] at $M = 2$) is exhausted by the engine (15654 nodes, independently in both engine implementations), and the $\gamma = 2\alpha$ branch is excluded by Beeson’s Theorem 11.18 [3] ($N \geq 45$ for that branch; his Table 3 lists and eliminates the unique $N = 44$ boundary candidate $(25, 11, 30)$). The third branch is the isosceles-base- β case of [2, Thm. 14]: at $(N, M) = (44, 6)$ the tiling equation gives $s = a/c = \frac{1}{2}$, hence the tile $(2, 3, 4)$ — angles $(\alpha, \beta, 2\alpha+\beta)$ with $3\alpha + 2\beta = \pi$, $\cos \alpha = \frac{7}{8}$, $\cos \beta = \frac{11}{16}$, $\cos \gamma = -\frac{1}{4}$ — and the target $(16, 16, 22)$ with base angles β . Here the exhaustive search does not exhaust: *it finds a tiling*, at node 858163.

Theorem 31. *A triangle can be cut into 44 congruent triangles: the isosceles triangle with sides $(16, 16, 22)$ is tiled by 44 congruent copies of the $(2, 3, 4)$ triangle (Fig. 1). Consequently $44m^2$ is a tile count for every $m \geq 1$.*

Proof. The tiling is exhibited explicitly (`code/engine/tiling_N44B.txt`: exact coordinates in $\mathbb{Q}(\sqrt{15})$), and its certificate is *machine-verified in Lean 4 with no axioms whatsoever* (`lean/Tiling44.lean`: the kernel checks, by pure computation over $\mathbb{Z}[\sqrt{15}]$ after scaling by 8, that all 44 triangles have squared side multiset exactly $\{256, 576, 1024\} = (8 \cdot (2, 3, 4))^2$, are positively oriented, lie in the closed target via half-plane checks, admit an explicit separating edge-line for each of the 946

pairs, and have signed areas summing exactly to the target's; `#print axioms` reports the theorem depends on no axioms). The bridge from the certificate to the tiling statement is classical: closed half-plane separation gives pairwise disjoint interiors (convexity), the vertices-in-target checks give containment (convexity of the target), and then exact equality of total area forces the union of the closed tiles to be the whole target, since the complement is open with measure zero. So this is an N -tiling in the strict sense — *unconditionally*: this theorem depends on none of the cited classification inputs. The consequence follows by subdividing each tile into m^2 similar copies. $\square$

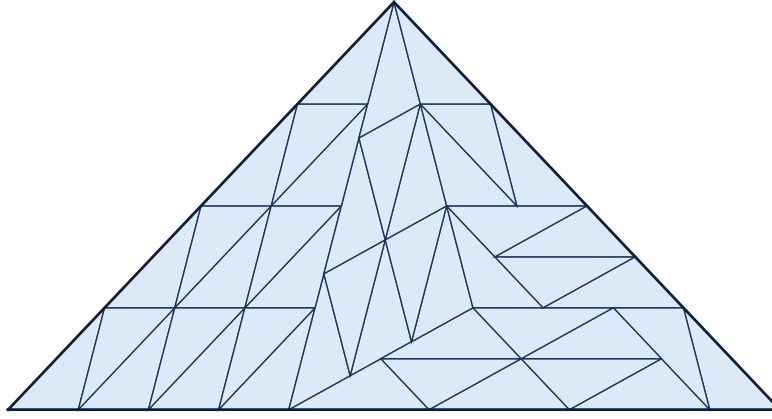


FIGURE 1. The 44-tiling of $(16, 16, 22)$ by the tile $(2, 3, 4)$, found by exhaustive search and verified exactly. To our knowledge the first tiling in any of the three isosceles $3\alpha + 2\beta = \pi$ cases, and the smallest known triangle tiling in any incommensurable branch (the previously smallest was Herdt's 1215).

Three remarks. First, 44 is not a square, not a sum of two squares ($11 \equiv 3 \pmod{4}$), and not 2, 3, 6 times a square: it is the first tile count *outside the commensurable families* to be certified realizable at small size, and the first new member of the tile-count set discovered by these methods — every other frontier value was excluded. Second, Beeson's constructions in [2] cover the two scalene $3\alpha + 2\beta = \pi$ shapes (his triquadratic tilings, smallest $N = 28$ — which, we note, settles 28's membership and is used in our accounting below); the isosceles cases had, to our knowledge, no known tiling, and the smallest known tiling in any incommensurable branch was Herdt's 1215. Third, the value demonstrates that the small members of the $3\alpha + 2\beta = \pi$ tables are not uniformly impossible: each must be searched, and Theorem 36's per-instance decision is genuinely two-sided.

The values 51, 55, 57, 62, and the $\gamma = 2\alpha$ boundary algorithm. The sweep continues past the forties. The values $51 = 3 \cdot 17$, $55 = 5 \cdot 11$ and $57 = 3 \cdot 19$ are squarefree, so the $\gamma = 2\alpha$ branch is closed by [3, Thm. 11.7], and every branch except isosceles- $(\alpha + \beta)$ dies arithmetically; each surviving instance is uniquely determined and the engine exhausts all six (62 dies in every branch outright, as 42 did):

instance	tile	target	nodes
$N=51, M=1$	(650, 51, 676)	(1326, 1326, 1275)	3649
$N=51, M=3$	(70, 51, 100)	(510, 510, 357)	180
$N=55, M=1$	(756, 55, 784)	(1540, 1540, 1485)	5911
$N=55, M=5$	(24, 55, 64)	(440, 440, 165)	40
$N=57, M=1$	(812, 57, 841)	(1653, 1653, 1596)	8961
$N=57, M=3$	(88, 57, 121)	(627, 627, 456)	193
$N=63, M=1$	(992, 63, 1024)	(2016, 2016, 1953)	15493
$N=63, M=3$	(12, 7, 16)	(84, 84, 63)	44859
$N=63, M=7$	(8, 63, 64)	(504, 504, 63)	2
$N=69, M=1$	(1190, 69, 1225)	(2415, 2415, 2346)	37995
$N=69, M=3$	(130, 69, 169)	(897, 897, 690)	523

The three $N=63$ rows close its isosceles- $(\alpha+\beta)$ branch. The value 63 itself then hangs on its $\gamma = 2\alpha$ instance — the boundary algorithm (below) leaves exactly one candidate, tile (9, 7, 12) with target (63, 63, 84) — and the engine exhausts it too, at 884921 nodes; to our knowledge the first value decided *inside* Beeson’s “swath of ignorance” [45, 1125] for the $\gamma = 2\alpha$ branch [3, §11.6]. The values 70 and 78 die in every branch outright, as 42 and 62 did. The value 66 reduces to its $\pi/3$ -equilateral instance (tile (11, 96, 91), side 264, coloring number $M = 4$), which Beeson rejects [5, Thm. 7]: the tiling must witness a relation $91j = 11\ell + 96m$ with $91j < 264$, and neither $j = 1$ nor $j = 2$ admits one. And 77 goes the *other* way: its $(2\alpha, \alpha, 2\beta)$ instance satisfies Beeson’s second tiling equation at $(M, s) = (5, \frac{1}{2})$ — $25 \cdot \frac{(2-\frac{1}{4})(3-\frac{1}{4})}{(1-\frac{1}{2})^2(2+\frac{1}{2})^2} = 77$ — and his four-component construction [2] realizes every solution, so *a triangle can be cut into 77 congruent triangles*: the $(2\alpha, \alpha, 2\beta)$ triangle over the same tile (2, 3, 4) that Theorem 31 uses for 44.

For values that are *not* squarefree the $\gamma = 2\alpha$ branch needs more than Theorem 11.7. Beeson’s isosceles paper provides exactly the needed machinery: the tile must be $(k^2, m^2 - k^2, mk)$ with $\gcd(k, m) = 1$, $2k > m > k$ [3, Lem. 11.2]; m is bounded in terms of N [3, Lem. 11.12]; the side X and base Y satisfy $X^2 = Nab$, $Y = (m/k)X$ and must decompose as $X = pa + qb + rc$ with $q, r \geq 1$ and $p + q + r < (N+1)/4$ [3, Lems. 11.11, 11.13] and $Y = ua + vb + wc$ with $v, w \geq 1$ (the base version of Lemma 11.13, whose pigeonhole proof is stated for the base); and the pair is rejected when every X -representation has $r = 1$ while every Y -representation has $w \in \{1, 2\}$ [3, Lem. 11.17]. We implemented this boundary algorithm independently (`code/gamma2alpha.py`) and calibrated it against Beeson’s reported results: it excludes 20, 28 and 44 and leaves 45 surviving via the tile (4, 5, 6), exactly as in [3, Thm. 11.18]; at 36 our filter is strictly weaker than Beeson’s direct argument (it reports a surviving boundary representation), so we use it only in the sound direction. Running it on the undetermined values: *the $\gamma = 2\alpha$ branches at $N = 56$ and $N = 60$ are empty* — no possible boundary tiling exists.

Consequently $N = 56$ reduces far: $\gamma = 2\alpha$ is closed by the algorithm, both isosceles- $(\alpha+\beta)$ instances are exhausted by the engine ($M=2$: tile (195, 56, 225), target (840, 840, 728), 853 nodes; $M=4$: tile (45, 56, 81), target (504, 504, 280), 89 nodes), and every other branch dies arithmetically — *except* the $2\pi/3$ -equilateral instance: tile (8, 7, 13), equilateral target of side 56, from the factorization $(s, t) = (12, 14)$ of Proposition 27. The membership of 56 is thus equivalent to that single equilateral instance — and the engine *exhausts it*: no tiling exists, at 156646 nodes, identically in both independent engine implementations (node-for-node agreement). Likewise $N = 60$ reduces to its equilateral instance alone (tile (5, 3, 7), side 30): its two isosceles- $(\alpha+\beta)$ instances are exhausted as well ($M=2$: tile (56, 15, 64), target (240, 240, 210), 123003 nodes; $M=6$: tile (4, 15, 16), target (120, 120, 30), 39 nodes) — and the equilateral instance itself is exhausted at 853731 nodes: $N = 60$ is excluded. (This also settles the $m=2$ member $X = 30$

of Zhang’s equiconstructibility family for $(5, 3, 7)$ by exact arithmetic; Beeson had reported $X < 105$ unrealizable for this tile in unpublished floating-point work.)

Theorem 32. *No triangle can be cut into 51, 55, 56, 57, 60, 62, 63, 69, or 78 congruent triangles.*

Proof. The branch sweeps and the exhaustive searches above; dependencies as in Theorem 29, together with [3, Thm. 11.7] for the $\gamma = 2\alpha$ branch at the squarefree values 51, 55, 57, 69; the boundary algorithm (Lemmas 11.2–11.17 of [3], `code/gamma2alpha.py`) for the $\gamma = 2\alpha$ branch at 56 and 60; the boundary algorithm’s single surviving candidate plus its 884921-node exhaustion at 63. (The value 66, previously listed here, is pending: Remark 4.) $\square$

Remark 33 (The value 70 — retracted to pending). An earlier version of this theorem included $N = 70$. Its single surviving instance is isosceles- α : tile $(6, 5, 9)$, $M = 8$, target $(45, 45, 70)$, which passes the tiling equation *and* side-integrality ($X = 45$, and indeed $(2f + e) = 8 \mid M$); the exclusion rested solely on $g \mid M$ from [2, Thm. 19] ($g = 3 \nmid 8$), whose proof is unsound (Remark 2); Beeson’s own Table 5 marks this very instance “?”. We therefore regard 70 as *undetermined* pending the direct exhaustive search of the instance, which is in progress. The value 21, whose analogous isosceles- α candidate (tile $(2, 3, 4)$, $M = 5$, target $(12, 12, 21)$) was likewise previously killed only by the unsound $g \mid M$, is excluded: the engine exhausts the instance (13 659 nodes, no tiling), so Theorem 30 stands with its dependence on [2, Thm. 19] removed. The same applies to $N = 39$, whose base- β candidate (tile $(12, 7, 16)$ on $(64, 64, 117)$) the engine exhausts in 825 724 nodes: Theorem 30 therefore stands for 39 as well, independently of [2, Thm. 14]. Of the values whose exclusion rested on an unsound divisibility, only 66 and 70 remain pending (Remark 4).

Remark 34. The $2\pi/3$ -equilateral exhaustions at $N = 56$ (tile $(8, 7, 13)$, side 56) and $N = 60$ (tile $(5, 3, 7)$, side 30) lie strictly inside the range $12 < N < 1215$ that Beeson leaves *open* for this case — he proves only $N \geq 12$ [5, §5] — and, cross-checked to the node in two independent engine implementations (156 646 and 853 731 nodes respectively), are to our knowledge the first rigorous exclusions inside it. The value 56 moreover settles the $m=1$ case of Zhang’s equiconstructibility family for $(8, 7, 13)$: the side-56 equilateral is *not* tileable by $(8, 7, 13)$, exactly as Zhang’s Conjecture 2 predicts (his construction realizes $X = 56m$ only for $m \geq 9$). It also closes the $m=1$ route of his Proposition 9(1) to the F_1 instance $N = 105$: any 105-tiling of $(105, 56, 91)$, if one exists, does not factor through an equilateral 56-block.

The values 76 to 80. The same branch sweep (`verify_frontier.py`) extends the determined initial segment past 75. For $N = 76 = 4 \cdot 19$ the only live branches are $\gamma = 2\alpha$ (excluded by the boundary algorithm of Lemmas 11.2–11.17 of [3], `code/gamma2alpha.py`) and the $3\alpha + 2\beta = \pi$ isosceles- $(\alpha + \beta)$ target with tile $(90, 19, 100)$ on the isosceles $(380, 380, 342)$, which the engine exhausts (955 474 nodes); every other branch is dead by its N -form. Hence 76 is not realizable. The prime $79 \equiv 3 \pmod{4}$ is excluded by Theorem 1; and 77, 80 are realizable — 77 by Beeson’s four-component construction [2], and $80 = 4^2 + 8^2$ by the biquadratic (commensurable) tiling [4].

Theorem 35. *No triangle can be cut into 76 congruent triangles. Every $N \leq 80$ except $N = 70$ (Remark 33) is thereby determined: the realizable values are exactly 1–13’s known members together with 44, 77, 80 and the standard commensurable and prime $\equiv 1 \pmod{4}$ values, and all others are excluded; in particular the contiguous record is every $N \leq 69$, extending through 80 once the pending 70-instance search completes.*

Proof. The branch sweep and $\gamma = 2\alpha$ boundary algorithm above settle 76; Theorem 1 excludes the prime 79; Theorems 30–32 give the composite exclusions through 78 other than 70; and the four-component and biquadratic constructions realize 77 and 80. $\square$

Decidability. The pattern of the last two subsections is not an accident. Every branch of the classification now reduces, for each fixed N , to a *finite* list of fully determined instances: the commensurable, similar-tile, right-angle and $\gamma = 2\alpha$ branches are decided by closed N -forms; in the sporadic $2\pi/3$ shapes the divisor structure of N (Theorems 20, 22, Proposition 26) leaves finitely many (tile, scale) pairs; the equilateral cases pin the tile from finitely many factorizations (Proposition 27) or coloring numbers ([5, Thm. 2], which determines the tile from (N, M)); and the five $3\alpha + 2\beta = \pi$ shapes have finitely many (M, s) candidates by their tiling equations [2]. Each surviving instance is a bounded existence question, and the corner-anchored search of `code/engine/` decides it in finitely many steps (the branching is provably complete and the depth is N). Hence:

Theorem 36 (Decidability). *It is decidable, given N , whether some triangle can be cut into N congruent triangles (under the same cited inputs as Theorem 1).*

This extends the decidability that Beeson established for equilateral targets [5, Thm. 4] to the whole problem. What decidability does *not* give is a closed-form description of the set. After Theorem 23 and Proposition 28, the necessary side is closed in *every* branch — the admissible values are Zhang’s families in the sporadic shapes and explicit divisor-instances in the equilateral cases — so the entire residual openness of the problem is a single question: *which admissible instances are realized*. Zhang’s trapezoid constructions realize the sporadic families for $m \geq 9$ in residue classes, and Laczkovich’s elliptic-curve analysis [11] governs equilateral realizability; the open kernel is the small members that are not already realized commensurably — sharpest $N = 105$ on the F_1 target of $(8, 7, 13)$ — and each such instance is individually decidable (Theorem 36).

Remark 37 (no invariant decides realizability). The directional invariants are *conserved* by the tiling process: each tile carries value $\pm(c + a - b)$ for f_α (resp. $\pm(c + b - a)$ for f_β), so placing one tile changes Φ_f of the unfilled region by $\pm V$ and the number of remaining tiles by one — in step, so that Φ_f/V stays integral and keeps the parity of the tile count throughout. Hence the only conditions such an invariant imposes are the admissibility conditions of Theorem 23, and *no* additive boundary invariant can distinguish an admissible target from a partially tiled one; none can obstruct an admissible instance, and none prunes the search below its combinatorial size. For the sharpest open instance $N = 105$ we confirmed this computationally: searching all directional weights of period ≤ 4 in the α -winding (reflections included), the only invariants are f_α ($M_\alpha = 5$) and f_β ($M_\beta = 21$), both admissible; and the vertex-type balance forced by the angle relations $a\alpha + b\beta + c\gamma \in \{\pi, 2\pi\}$ admits 42,282 nonnegative integer solutions. Deciding realizability of an admissible instance therefore lies beyond every linear or coloring invariant; it is a genuinely combinatorial question, which is why the search of Theorem 36 appears to be essential and why Zhang’s conjecture is hard at its sharpest point.

Remark 38 (the tiling group collapses as well). The strongest standard combinatorial obstruction is the Conway–Lagarias *tiling group* $T = \langle \chi_v \rangle / \langle \text{collinear additivity; tile boundary words} = 1 \text{ in every orientation and reflection} \rangle$; a tileable region R has $\chi(\partial R) = 1$ in T . Writing edge vectors as complex numbers in $K = \mathbb{Q}(\sqrt{3}, i)$ (rotation by α is multiplication by $\zeta_\alpha = (11 + 4\sqrt{3}i)/13$, by $\frac{\pi}{3}$ by $(1 + \sqrt{3}i)/2$, reflection by conjugation) and reducing to finite quotients $\mathbb{F}_{p^2}$ with $p \equiv 11 \pmod{12}$ — so $\sqrt{3} \in \mathbb{F}_p$ but $i \notin \mathbb{F}_p$ and the plane stays genuinely two-dimensional — the tile relators, *together with* the commutators [class-1(tile), generator] that normal closure forces (a tile word’s class-1 image is itself a relator, hence not central), fill the entire commutator layer. So T collapses to its rank-two abelianization (M_α, M_β) : the class-2 quotient is trivial while the class-1 layer keeps rank 2. As $N = 105$ is admissible, $\chi(\partial_{105}) = 1$, and the tiling group does not obstruct it. We verified this exactly for $p \in \{23, 47, 59, 71, 83, 107, 131\}$ (`verify_tiling_group.py`), and

— extending to the free nilpotent quotient through class 4 and to a range of finite non-abelian groups (through S_5 , $\mathrm{GL}_2(\mathbb{F}_p)$, $\mathrm{PSL}(2, 7)$ and the relevant Frobenius groups, which carry genuine non-abelian data: sensitivity controls fire on 92–96% of tile-consistent representations) — found *no* tile-consistent quotient sending ∂_{105} off the identity, with the reptile controls $N = 4, 9$ trivial throughout and a Φ -nonadmissible loop not. The no-go of Remark 37 therefore extends past the linear invariants to the full tiling group: on this tile the tiling-group obstruction is exactly Φ , so deciding realizability of $N = 105$ lies outside every standard tiling invariant.

State of the full problem. Assembling: the commensurable branch is completely determined (squares, sums of two squares, and 2, 3, 6 times squares, all realized [4]); the similar-tile and right-angle branches are settled in [1, 12]; the primes are settled by Theorem 19. What remains open for the full determination of the tile-count set is: sufficiency in the sporadic $2\pi/3$ branches (Zhang’s completeness conjecture [7]; the admissible spectra above are the outer bounds), the $3\alpha + 2\beta = \pi$ and $\gamma = 2\alpha$ branches beyond the partial results of [2, 3], and the equilateral targets for incommensurable tiles, where Laczkovich has connected the possible N to rational points on elliptic curves [11]. With 7 and 11 excluded by [4], the primes by Theorem 19, 14, 15, 21, 22, 30, 33, 35, 38, 39, 42, 46 by Theorems 29 and 30, 51, 55, 56, 57, 60, 62, 63, 69, 78 by Theorem 32 (66 and 70 pending, Remarks 4 and 33), 44 *realizable* by Theorem 31, 77 realized by Beeson’s four-component construction [2], 28 realized by his triquadratic tilings [2], and the remaining values commensurably realized (45, 48, 49, 50, 52, 53, 54, 58, 61, 64, 65, 68, 72, 73, 74, 75) or prime (47, 53, 59, 67, 71), the membership of *every* $N \leq 75$ is now determined. The value $N = 76$ is likewise excluded ($\gamma = 2\alpha$ empty by the boundary algorithm; its one isosceles- $(\alpha + \beta)$ instance, tile (90, 19, 100) on (380, 380, 342), exhausted by the engine at 955 474 nodes); with 79 prime (Theorem 1) and $80 = 4^2 + 8^2$ realized, *every* $N \leq 80$ *except the pending* 70 is now determined (Theorem 35).

Standing of the inputs. For the record, the external inputs to Theorem 1 and their publication status: Laczkovich’s classification papers [9, 10] and the rep-tiling theorem of Snover–Waiveris–Williams [12] are refereed journal articles, and Beeson’s isosceles paper [3] is accepted (to appear in *Integers*); Beeson’s remaining branch papers [1, 2, 4, 5] and the Beeson–Zhang rationality theorem [6], which is load-bearing for the integrality of the tile, are arXiv preprints at the time of writing. Zhang’s constructions [7] are cited for context only and carry no logical weight here. The parts original to this paper (Sections 3–5 and 7, together with the shape enumeration of Section 2) are self-contained given those inputs, with the arithmetic layer machine-checked (Section 8).

8. VERIFICATION

The finite and numerical claims are checked by the scripts in the repository (exact arithmetic). `verify_shapes.py` reproduces the eleven-shape enumeration of Section 2 and the values N_0 of Section 3. `verify_invariant.py` confirms Lemma 13 ($C_f(t) = \pm(c + a - b)$ over all orientations of T), Lemma 12 (both sides agree on explicit subdivision tilings, and (6) cancels at a non-edge-to-edge incidence), and Lemma 15 (no counterexample among primitive 120° -triples up to $c \approx 9 \times 10^6$). `verify_spectrum.py` checks Section 7: the two-positive-squares decomposition for all primes $p \not\equiv 3 \pmod{4}$ below 10^5 ; the scale structure $b \mid k^2 \iff de \mid k$; the admissible spectrum of Theorem 20 (no prime is admissible; 33 and 46 are); and that every instance of $e \mid (a + b - c)$ with $a, b < 3000$ arises from the j -classification of Proposition 21, with none for $d = 1$. As a positive control, both necessary conditions of Theorem 16 hold on the genuine 2673-tile isosceles tiling of Herdt [3]: for its tile (5, 3, 7) and scale $k = 27$, $N = k^2(a + 2b)/b = 2673$ and $M = k(2b + a - 2c)/(c + a - b) = -9$ are integers, as they must be on any true tiling.

The entire *arithmetic and combinatorial* layer of the proof is formalized and machine-checked in Lean 4 with Mathlib (`lean/Erdos634.lean`, twenty-three theorems); every theorem depends only on the standard axioms `propext`, `Classical.choice`, `Quot.sound`, with no `sorry`:

`shape_enumeration`: the eleven-triple completeness of Section 2 — realizable corner types with $\sum m_i = 0$, $\sum k_i = 3$ are exactly the eleven listed — as pure integer arithmetic;
`k_not_dvd_sum_sub`, `M_not_int`: Lemma 15 ($((c - a - b)/\sqrt{b}) \notin \mathbb{Z}$);
the full arithmetic of Theorem 16 in one master statement: no prime N satisfies the 120° -relation, the area equation $Nb = k^2(a + 2b)$, and the integrality $(c + a - b) \mid k(2b + a - 2c)$ simultaneously (`iso_reduction_identity`, `prime_count_forces_scale`, `no_prime_isosceles_count`);
`add_not_prime`, `not_prime_of_two_le`, `F1_count_not_prime`–`F4_count_not_prime`: the scalene counts of Proposition 11 are composite ($a + b$ is never prime for a primitive 120° -triple, and the F_2, F_3, F_4 counts factor);
`prime_three_mod_four_excluded`: a prime $p \equiv 3 \pmod{4}$ with $p > 3$ is neither a square, a sum of two squares, nor $2n^2$, $3n^2$ or $6n^2$ (the commensurable branch of Section 2), via Fermat’s two-squares theorem;
`prime_sum_two_pos_squares`: a prime $p \not\equiv 3 \pmod{4}$ is a sum of two *positive* squares (the achievability half of Theorem 19);
`iso_admissible`: Theorem 20 — given $b = de^2$ with d squarefree, the area equation and Φ -divisibility force $k = dew$, $N = dw^2(a + 2b)$ and $e \mid w(c - a - b)$;
the finite arithmetic of the frontier sweeps, in eight theorems: the $\pi/3$ -equilateral square criterion fails at 14 and 15; the first tiling equation has no admissible solution there; the two equilateral factor-pair uniqueness statements; the boundary congruences that kill the isosceles candidates at 14 and 22; the F_1 invariant kill at 21; and the parity kill of 46.

What is *not* formalized is exactly the geometry: the direction grid of Section 2 and Lemmas 12 and 13, which are proved in the text and checked numerically; Mathlib has no theory of planar dissections.

ACKNOWLEDGEMENTS

This work was carried out with substantial assistance from an artificial-intelligence system (Anthropic’s Claude), which proposed the signed-direction invariant, performed the symbolic and numerical verification, found the elementary proof of Lemma 15, and produced the Lean formalization, under the author’s direction and review. The argument depends on the cited classification of Laczkovich and Beeson for the branches other than the $2\pi/3$ tile, and on the rationality theorem of Beeson and Zhang; the shape list coincides with Zhang’s six sporadic types [7]. The present contribution is the invariant of Section 4, the consequent exclusion of *every* prime for the isosceles $2\pi/3$ branch (where Beeson had ruled out only $N < 36$ and left primality open), and the self-contained reduction of the scalene types. Its arithmetic layer is machine-checked (Section 8); the two geometric lemmas rest on written proofs supported by numerical checks, and the whole is offered for independent refereeing.

REFERENCES

- [1] M. Beeson, *Triangle tiling I: The tile is similar to ABC or has a right angle*, arXiv:1206.2231.
- [2] M. Beeson, *Triangle tiling III: the case $3\alpha + 2\beta = \pi$* , arXiv:1206.2229.
- [3] M. Beeson, *Tilings of an isosceles triangle*, arXiv:1206.1974; to appear in *Integers*.
- [4] M. Beeson, *No triangle can be cut into seven congruent triangles*, arXiv:1811.09723.
- [5] M. Beeson, *Tiling an equilateral triangle*, arXiv:1812.07014.
- [6] M. Beeson and Y. X. Zhang, *Rationality of certain triangle tilings*, arXiv:2604.01314.

- [7] Y. X. Zhang, *Tiling triangles with $2\pi/3$ angles*, arXiv:2512.22696.
- [8] M. Beeson, M. Laczkovich and Y. X. Zhang, *Solution of Erdős Problem 633*, arXiv:2604.03609.
- [9] M. Laczkovich, *Tilings of triangles*, Discrete Math. **140** (1995), 79–94.
- [10] M. Laczkovich, *Tilings of convex polygons with congruent triangles*, Discrete Comput. Geom. **48** (2012), 330–372.
- [11] M. Laczkovich, *Rational points of some elliptic curves related to the tilings of the equilateral triangle*, Discrete Comput. Geom. **64** (2020), 985–994.
- [12] S. L. Snover, C. Waiveris and J. K. Williams, *Rep-tiling for triangles*, Discrete Math. **91** (1991), 193–200.
- [13] A. Soifer, *How does one cut a triangle?*, 2nd ed., Springer, 2009.